

# Manual

## Controlling of Articles/Scrap BDE-CAA 8.1

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# 1 Übersicht Controlling Artikel / Ausschuss

## Purpose

The overview controlling article / scrap function package offers the ability to evaluate data entered as a part of shop floor data collection (BDE). You use the function package if:

- You would like to evaluate the shop floor data entered relating to articles
- You would like to evaluate the scrap data entered.

## Integration

The evaluations are based on the data collected during production. For this purpose, the article-based evaluations aggregate the quantities and times relating to an order, while the scrap evaluations analyze the scrap quantities recorded.

## Features

- Article profile
  - Article profile with comparison to the scrap accumulated as well as the time recorded (consolidated in resource performance accounts)
- Scrap statistics
  - Workplace and article-related scrap (reason) statistics. Tabular display of total and scrap quantity, scrap rate and the calculated scrap ratio. Graphic display of scrap quantities by workplace, article or scrap reason.
  - Scrap statistics expanded with columns like machine, machine designation, cost center, group, article, order, tool, scrap reason, scrap quantity, scrap rate as % of total quantity (formula), total quantity, scrap quantity as % of total scrap quantity
  - Ability to select by machine, order, article number, cost center, group, tool, time and shift
  - Optional calculation of scrap ratios as % of total quantity, of articles or machines
  - Display of yield, rework, problem quantity and total quantity
- Scrap profile
  - Evaluations of scrap quantities entered over time. Staging of data in table and pivot table form. Data evaluation in pivot table form by various criteria. Graphical presentation of data in pivot table form.

## 2 Article Profile

### Summary

|                        |                                                        |
|------------------------|--------------------------------------------------------|
| Menu                   | Order Management → Order Controlling → Article Profile |
| Transaction code       | artpf                                                  |
| Function authorization | artpf                                                  |

The Article profile application is designed for users in the departments Production controlling, Order controlling or Final costing. It compares orders across a defined period of time that produced the same article and displays comparisons of their production times and downtimes. Based on the selection made via the order header article (the order's final product) and, optionally, a defined period of time, all orders that produced that article are displayed along with their results.

### Integration

The application is based on the order-related quantities and times entered during production.

### Requirement

The application only looks at production orders - not individual operations. Only those production orders are considered that have the control status "finished" in their order headers. Production orders with a different control status are not considered.

### Selection criteria

The application provides the selection criteria listed below: Irrespective of these selection criteria, generally only orders are selected that have already started (in process) or that are finished.

#### Final product

This selection criterion refers to the article in the order header. All orders are displayed, to which the article entered has been assigned. The use of wildcards (placeholders \*) is allowed.

#### Article designation

This selection criterion refers to the article designation defined in the order. All orders are displayed that contain the article designation selected.

#### Order status

This selection criterion refers to the status of the order. Only those orders are displayed with an order status that matches the criteria entered. Irrespective of this restriction, generally only orders are considered and selected that have already started (in process) or that are finished.

**Category**

This selection criterion refers to the order type category of the order. Only those orders are displayed with an order type that is associated with the category entered.

**Order**

This selection criterion refers to the order number. Orders are displayed that contain the order number entered. There is an option to use wildcards (placeholders \*).

**Order type**

This selection criterion refers to the order type. Orders of the order type entered are displayed.

**Order end ... until ...**

This selection criterion refers to the point in time at which the order(header) is changed to "finished" status. In this case, only the finished orders are displayed with an actual end that is set between the selected dates. This occurs irrespective of whether there is a restriction on the order status.



The responsibility area is not checked in this application.

**Article profile detail application (table)**

All orders that match the selections entered are displayed in the article profile detail application table view. A selection of columns is described below:

**Status**

Current status of the order. Generally, only orders are considered that have started or that are finished.

**Status since date, status since time**

Point in time as of which the current status of the order applies.

**Order**

Order number of the order

**Final product**

Article number of the order(header).

**Article designation**

Name of the article listed in the order(header).

**SUT, DCI, SCI, LCI, IMS, IMN, SET, STA, U8, U9, MUT**

The durations that were posted to each resource performance account are displayed in the columns.

**Retention period of order**

The retention period of the order is the period between the time the order was first transferred from the PPS system ("order release" = order header creation date in HYDRA) and the time of the actual logoff (in terms of time) of the last active operation of the order.

Please note: Whether the time of the order transfer is the time it was first transferred from the PPS system or the time it was re-transferred because it had been deleted in the meantime for technical reasons, cannot be determined or taken into consideration using HYDRA. If an order is transferred more than once, and with each transfer the previous order is deleted, the creation date of the order header is the time of the last transfer.

Please note: If the order is not yet finished, the value calculated here is not significant.

**Lead time**

The order duration is the period between the time the first operation of the order is logged on and the time the last active operation is logged off.

Please note: If the order is not yet finished, the value calculated here is not significant.

**Processing time**

The processing time of the order is the sum of the main utilization times (RPA 11) of all recordable active operations.

**Downtime period**

The downtime period of the order is the sum of the downtimes (RPA 1..6, RPA 8..10) of all recordable active operations.

Please note: the setup time can be found in the corresponding resource performance account (RPA 7/ SET).

**Assignment time**

The occupancy time is the sum of the setup time (RPA 7), processing time (RPA 11) and downtimes (RPA 1..6, RPA 8..10) of all recordable active operations.

**Yield**

Recorded yield in base quantity unit of the last recordable operation. This requires that the unit in which the quantity is entered (primary quantity unit) can be converted into the base quantity unit.

**Scrap**

Sum of the scrap quantities entered for all operations posted in base quantity unit.

**Rework**

Sum of the rework quantities entered for all operations posted in base quantity unit.

**Open quantity**

Sum of the recorded problem quantities entered for all operations posted in base quantity unit.

**Unit**

Quantity unit of the order

**Rate of capacity utilization**

The rate of capacity utilization is the ratio of processing time (RPA 11) to occupancy time (RPA 1..11)

**Setup rate**

The setup rate is the ratio of setup time (RPA 7) to occupancy time (RPA 1..11)

**Article profile detail application (graphic)**

By selecting one or more orders in the table, the following information is displayed in the graphic for those orders:

**Upper graphic**

In the upper graphic of the article profile, three bars are displayed for each of the orders selected in the table:

- On the left: Scrap in percent based on the finished total quantity of the order (yield + scrap + rework + problem quantity in base quantity unit).
- In the middle: Main utilization time in percent based on the occupancy time of the order
- On the right: Downtime periods (RPA 1 to 10) in percent based on the occupancy time ( RPA 1 to 11) of the order

The colors of the time bars correspond to the defined RPA colors; scrap bars are generally displayed in red.

**Lower graphic**

In the lower graphic of the article profile, the absolute values of a single RPA or of the scrap are displayed in bar form for each selected order. The user determines which value is displayed by clicking in the upper graphic.

The individual order numbers are displayed on the X axis, while the unit of the value is displayed on the Y axis. If durations are greater than 24 hours, they are displayed in days.hours:minutes:seconds.

Each value selected (scrap or name of the resource performance account) is displayed beneath the graphic.





## 3 Scrap Statistics

### Summary

|                        |                                                         |
|------------------------|---------------------------------------------------------|
| Menu                   | Order management → Order controlling → Scrap statistics |
| Transaction code       | scrst                                                   |
| Function authorization | scrst                                                   |

Anyone interested in improving scrap rates in a company must have knowledge of how and where scrap is accrued. The scrap statistics provide a tool that allows not only the scrap quantity to be reported, but also all of the scrap reasons that applied during a selected period. In it, scrap recorded automatically is considered in the evaluation. The scrap statistics use graphic illustrations to provide the user with impressive hit lists of the most frequent scrap reasons.

### Integration

The order-related log records for the record type "T" form the data basis used to display quantities in the scrap statistics. These are generated at the shop floor terminal either due to manual partial uploads or because of an operation interruption or logoff, provided that quantities are recorded in this case.

### Selection criteria

The application provides the following selection criteria:

#### **Workplace/ short designation/ group/ cost center/ company/ responsibility area**

You use this index tab to define which workplaces/ machines are used in the evaluation. In the process, data are accessed from the master data of the workplace/ machine configuration.

#### **Article/ article designation**

Selection by article or article designation

#### **Order/ order type**

The evaluation can be limited to an order or an order type.

#### **Date**

The time period entered restricts the selection by log records. The log records of record type T are selected that have a posting time within the defined period.

#### **Shift**

Within the entered period, only those log records are selected that are assigned to the shift entered.

#### **Time**

Within the entered period, only those log records are selected that were posted during the period entered.

### Reason

Selection based on a defined scrap reason.

### Sales order

If a sales order is assigned in the production order, the selection can be based on it.

### Project number

If a project number is assigned in the production order, the selection can be based on it.

### Report group

The report group option allows you to limit the selection to a specific group of workplaces/ machines. For this purpose, the report group must be defined in the [group configuration](#) option.

### Show yield

If this option is set, the yield recorded in the selection period is also shown.

## Scrap statistics detail application

The scrap statistics detail application generates a table view of all of the scrap reasons accrued over a selected period of time along with the scrap quantity. The order-related log records for the record type "T" form the data basis for the selection. The quantities calculated generally relate to the primary quantity unit. When these are summed up, the different quantity units are not converted.



In the scrap statistics, only operations are displayed that posted at least one partial upload with scrap during the evaluation period.

Please keep in mind that a yield in the column Yield (P) is then shown if the option "Show yield" is set. It is shown in a separate line per order/ OP and workplace.

Please note with regard to the selected columns:

### Scrap

The scrap column shows the scrap quantity relating to scrap reason, workplace and order/ operation.

Please keep in mind that the scrap quantity shown may also be negative in certain constellations. This can be the case, for example, if scrap is offset against yield.

### Scrap rate

Proportion of scrap (primary quantity unit) as compared to the total quantity produced of the operation (primary quantity unit) during the evaluation period.

The term total quantity is understood as the sum total of all four quantity accounts (yield, scrap, rework, open quantity). Please keep in mind that the rework quantity and open quantity are not shown in the scrap statistics.

Calculation:

$$\text{Scrap rate} = 100.0 * \text{EGR\_AUS} / (\text{ANR\_GUTP} + \text{ANR\_AUSP} + \text{ANR\_NCHP} + \text{ANR\_PRBP})$$

## **Comparison yield - scrap detail application**

This detail application provides a comparison of the yield and scrap quantity shown in the form of a bar chart. In it, the bars represent the relationship between the two quantity types. The percentage is visualized in each bar.

## **Scrap hit list (scrap reasons) detail application**

The scrap hit list (scrap reasons) is a compilation of the recorded scrap quantities, grouped by scrap reason and shown in graphic form.

The bar chart displayed depends on which entries were highlighted in the tabular detail application. If all of the entries shown in the table are to be taken into account in the chart, highlight the entire table by clicking on the box at the top left. We recommend that you deactivate the option "Show yield" when highlighting.

The bar chart shows the accumulated value per scrap reason for each bar over all selected entries as well as the percentage in relation to the entire scrap quantity determined during the evaluation period. The total scrap quantity is displayed in the totals line of the "scrap statistic" detail application.

Example: the detail application "scrap statistic shows a total scrap quantity of 1.171 in the totals line for a selected period. If an entry showing a scrap quantity of 122 is selected, the graphic shows a percentage of 14.42.

## **Scrap hit list (workplaces) detail application**

The scrap hit list (workplaces) is a compilation of the recorded scrap quantities, grouped by workplace/machine and shown in graphic form.

The bar chart displayed depends on which entries were highlighted in the tabular detail application. If all of the entries shown in the table are to be taken into account in the chart, highlight the entire table by clicking on the box on the top left. We recommend that you deactivate the option "Show yield" when highlighting.

The bar chart shows the accumulated value per scrap reason for each bar over all selected entries as well as the percentage in relation to the entire scrap quantity determined during the evaluation period. The total scrap quantity is displayed in the totals line of the "scrap statistic" detail application.

## Toolbar



### Order information

Calls up the [order information](#) for the currently selected order.

## 4 Scrap Profile

### Summary

|                        |                                                      |
|------------------------|------------------------------------------------------|
| Menu                   | Order Management → Order Controlling → Scrap Profile |
| Transaction code       | scrf                                                 |
| Function authorization | scrf                                                 |

Anyone interested in improving scrap rates in a company must have knowledge of how and where scrap is accrued. The scrap profile is a tool that allows not only the scrap quantity to be reported, but also all of the scrap reasons that applied during a selected period.

### Selection criteria

The application provides the following selection criteria:

#### Date

The time period entered restricts the selection by log records. The BDE log records of record type T are selected that have a posting time within the defined period.

#### Shift

Within the entered period, only those log records are selected that are assigned to the shift entered.

#### Time

Within the entered period, only those log records are selected that were posted during the period entered.

#### Workplace/ group/ cost center/ company/ short designation/ designation/ responsibility area

These entries define which workplaces/ machines are used in the evaluation. In the process, data is accessed from the master data of the workplace/ machine configuration.

#### Report group

The report group option allows you to limit the selection to a specific group of workplaces/ machines. For this purpose, the report group must be defined in the [Group configuration](#) option.

#### Reason

Selection based on a defined scrap reason.

#### Order/ order type

The evaluation can be limited to an order.

#### Article

Selection based on article.

**Sales order**

If a sales order is assigned in the production order, the selection can be based on it.

**Project number**

If a project number is assigned in the production order, the selection can be based on it.

**Scrap profile detail application**

The scrap profile detail application generates a table view of all of the scrap reasons accrued over a selected period of time along with the scrap quantity.

**Note about the column "Scrap rate":**

The scrap rate relates to the total quantity posted during the selected period of time (sum total of all yield, scrap, rework quantity and open quantity in the primary quantity unit) for the combination workplace/ MES order number.

Example: The following quantities are recorded at a workplace for an order/ operation for a defined period of time:

| Day | Yield | Scrap |
|-----|-------|-------|
| 1   | 10    | 1     |
| 2   | 12    | 0     |
| 3   | 0     | 0     |
| 4   | 14    | 1     |
| 5   | 8     | 2     |

If a selection is made over the period of time Day 2 - Day 5, on each of the days 4 and 5 scrap quantities accrued that are displayed in the scrap profile. The scrap rate relates to the quantity posted during the entire selection period (here:  $12+14+8+1+2 = 37$ ):

| Day | Scrap | Scrap rate |
|-----|-------|------------|
| 4   | 1     | 2.7        |
| 5   | 2     | 5.4        |



Because a scrap rate calculation also includes postings that show a yield, but with no scrap, there may also be entries with a scrap quantity of zero in the displayed results.

If necessary, these can be hidden using the Filter editor.

## Pivot table detail application

The pivot table detail application generates a table view of all of the scrap reasons accrued over a selected period of time along with the scrap quantity. For each scrap reason, for example, (depending on the pivot table settings) one column is displayed. The scrap recorded automatically is shown separately.

### Scrap (P)

This pivot element contains the primary quantities.

### Scrap reason

This pivot element contains the scrap reasons (number designated to the scrap reason).

### Date

This pivot element contains the date information for the accrued scrap.

In addition to the fields listed below, the detail application also contains a number of other fields. You can select these additional fields from the field list. To do so, with your mouse, right click in the area above the column headings and select the entry "Show field list" in the context menu.



Because a scrap rate calculation also includes postings that show a yield, but with no scrap, there may also be entries with a scrap quantity of zero in the displayed results.

If necessary, these can be hidden using the Filter editor.